Eigenvalues and inverse problems

Conditioning of a random Krylov compression of a cyclic shift

Difficulty: challenging **Importance:** interesting to specialist

Status: Solved

Independently reviewed resolution - 2026-09-11

Theorem 1 proves $\mathbb{E}\kappa_V(H_k) \le 17n^2k$ for the exact complex-sphere cyclic-shift model. Markov's inequality gives the uniform 0.99 target with $C = 1700$ and $c = 3$. Sections 5 and 6-7 give two probability proofs; real starts and arbitrary nonnormal inputs are outside the result.

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The difficulty, importance and rating rationale below are historical assessments of the original open target.

Rating rationale: Challenging reflects a least-singular-value bound for a dependent random Krylov matrix; specialist impact concerns a structured compression mechanism in eigenvalue algorithms.

Problem statement

Let $C_n \in \mathbb{C}^{n\times n}$ be the cyclic shift, $C_ne_j = e_{j+1}$ for $j < n$ and $C_ne_n = e_1$. Draw b uniformly from the complex unit sphere. For $2 \le k < n$, let Q have orthonormal columns spanning

$$\mathcal{K}_k(C_n, b) = \operatorname{span}\{b, C_nb, \dots, C_n^{k-1}b\}, \qquad H = Q^*C_nQ.$$

This subspace has dimension k almost surely. Define $\kappa_V(H) = \inf_{H=VDV^{-1},\ D \text{ diagonal}} \|V\|_2\|V^{-1}\|_2$, and set it to $+\infty$ if H is not diagonalizable. Do universal constants $C, c > 0$ exist such that, for every $n \ge 3$ and $2 \le k < n$, with the same constants,

$$\Pr\{\kappa_V(H) \le Cn^c\} \ge 0.99?$$

Changing the orthonormal basis of the same Krylov space does not affect this question. Randomness belongs to the starting vector; replacing Q by an independently Haar-distributed subspace would change the model.

Reference

Amsel et al., *Linear Systems and Eigenvalue Problems*, Problem 3.5, which explicitly singles out the circulant shift. This entry fixes "high probability" to a uniform 0.99 success target.

Earlier status check — 2026-09-08

Searches for *Krylov circulant shift condition 2026* and *random Krylov compression eigenvector condition number* found no solution. The deterministic starting vector e_1 gives a Jordan compression, so the unrandomized statement would be false; that exceptional example does not settle the probabilistic problem. The workshop report's 20 August update reports Peng's obstruction for arbitrary real diagonalizable inputs with ill-conditioned eigenvectors and real Gaussian starts. It explicitly leaves the normal-input case open, which includes the cyclic shift here.

Audit update — 2026-09-10

Rechecked workshop version 3, Problem 3.5 and its August update. The arbitrary-diagonalizable counterexample does not cover this normal cyclic shift. Random Krylov/circulant conditioning searches found no bound resolving the displayed structured case.